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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 26th July Evening Shift

JEE Main Question Paper with Answer Key

26 Jul 2022	90	25	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	19 questions
Mathematics	20 questions
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Out of syllabus	25 questions
Answer key	at the end

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Chemistry 19 questions

Q1. MCQ (Single Correct Answer)

Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in $3.3\ \text{g}$ of hemoglobin is

(Given: Atomic mass of Fe is $56\ \text{u}$, $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$.)

(A) 1.21×10^5 (B) 12.0×10^{16} (C) 1.21×10^{20} (D) 3.4×10^{22}

Open solution

Q2. MCQ (Single Correct Answer)

Arrange the following in increasing order of their covalent character.

A. CaF_2

B. $CaCl_2$

C. $CaBr_2$

D. CaI_2

Choose the correct answer from the options given below.

(A) B < A < C < D (B) A < B < C < D (C) A < B < D < C (D) A < C < B < D

Open solution

Q3. MCQ (Single Correct Answer)

Class XII students were asked to prepare one litre of buffer solution of $\text{pH}\ 8.26$ by their Chemistry teacher. The amount of ammonium chloride to be dissolved by the student in $0.2\ \text{M}$ ammonia solution to make one litre of the buffer is :

(Given: $\text{pK}_b\ (NH_3) = 4.74$

Molar mass of $NH_3 = 17\ \text{g mol}^{-1}$

Molar mass of $NH_4Cl = 53.5\ \text{g mol}^{-1}$)

(A) 53.5 g (B) 72.3 g (C) 107.0 g (D) 126.0 g

Open solution

Q4. MCQ (Single Correct Answer)

At 30°C, the half life for the decomposition of AB_2 is $200\ \text{s}$ and is independent of the initial concentration of AB_2. The time required for 80% of the AB_2 to decompose is

Given: $\log 2 = 0.30 \quad \log 3 = 0.48$

(A) 200 s (B) 323 s (C) 467 s (D) 532 s

Open solution

Q5. MCQ (Single Correct Answer)

The metal complex that is diamagnetic is (Atomic number: $\text{Fe}, 26; \text{Cu}, 29$)

(A) $K_3[Cu(CN)_4]$ (B) $K_2[Cu(CN)_4]$

(C) $K_3[Fe(CN)_4]$ (D) $K_4[FeCl_6]$

Open solution

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Q6. MCQ (Single Correct Answer)

The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

Open solution

(A) $-\mathrm{COOH} > -\mathrm{CONH_2} > -\mathrm{COCl} > -\mathrm{CHO}$

(B) $-\mathrm{SO_3H} > -\mathrm{COCl} > -\mathrm{CONH_2} > -\mathrm{CN}$

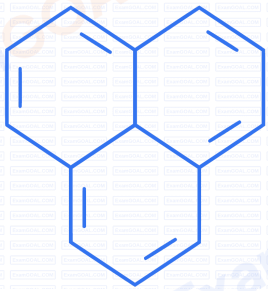
(C) $-\mathrm{COOR} > -\mathrm{COCl} > -\mathrm{NH_2} > \mathrm{C{=}O}$

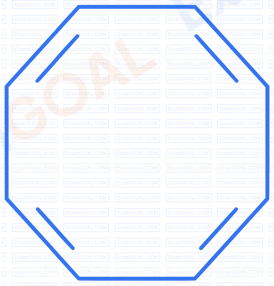
(D) $-\mathrm{COOH} > -\mathrm{COOR} > -\mathrm{CONH_2} > -\mathrm{COCl}$

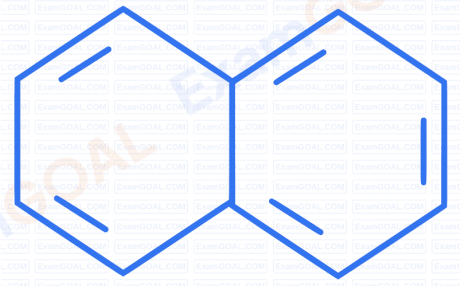
Q7. MCQ (More than One Correct Answer)

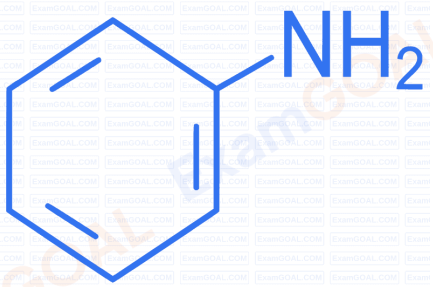
Which of the following is not an example of benzenoid compound?

Open solution

(A) 

(B) 

(C) 

(D) 

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Q8. MCQ (Single Correct Answer)

Hydrolysis of which compound will give carbolic acid?

(A) Cumene

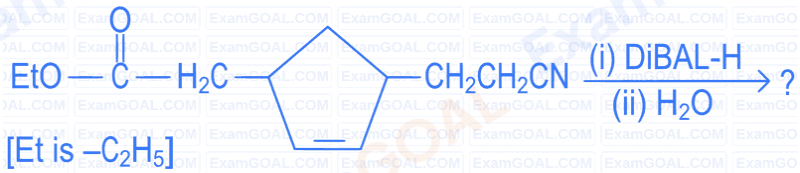
(B) Benzenediazonium chloride

(C) Benzal chloride

(D) Ethylene glycol ketal

Open solution

Q9. MCQ (Single Correct Answer)



[Et is $-C_2H_5$]

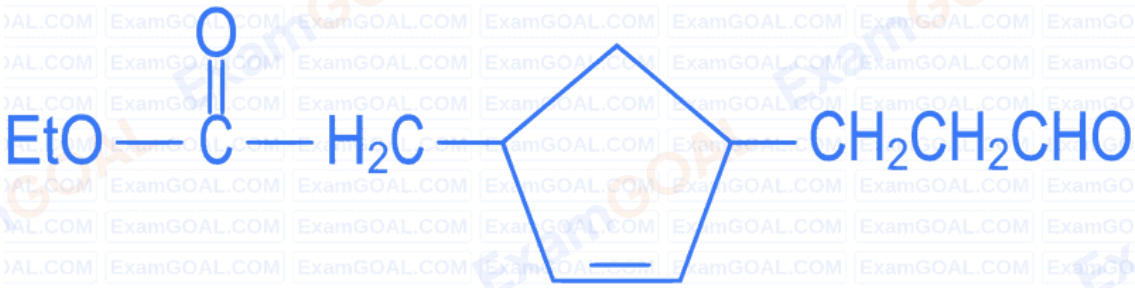
Open solution

Consider the above reaction and predict the major product.

(A)



(B)



(C)



(D)

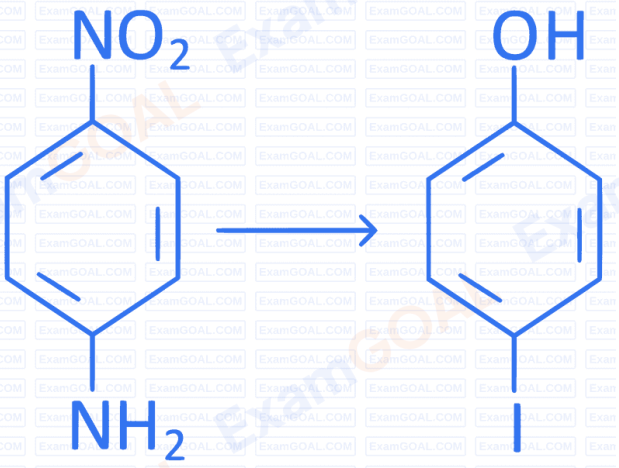


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Q10. MCQ (Single Correct Answer)

Open solution

The correct sequential order of the reagents for the given reaction is



(A) $HNO_2, Fe/H^+, HNO_2, KI, H_2O/H^+$

(B) $HNO_2, KI, Fe/H^+, HNO_2, H_2O/$ warm

(C) $HNO_2, KI, HNO_2, Fe/H^+, H_2O/H^+$

(D) $HNO_2, Fe/H^+, KI, HNO_2, H_2O/$ warm

Q11. MCQ (Single Correct Answer)

Open solution

Animal starch is the other name of

(A) amylose. (B) maltose. (C) glycogen. (D) amylopectin.

Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A: Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

Reason R: Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

(C) A is true but R is false.

(D) A is false but R is true.

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Q13. Numerical

Consider an imaginary ion $^{48}_{22}X^{3-}$. The nucleus contains 'a'% more neutrons than the number of electrons in the ion. The value of 'a' is _____. [nearest integer]

Open solution

Q14. Numerical

For the reaction

$H_2F_2(g) \rightarrow H_2(g) + F_2(g)$

$\Delta U = -59.6\ kJ\ mol^{-1}$ at $27^\circ C$.

The enthalpy change for the above reaction is $(-)$ _____ $kJ\ mol^{-1}$ [nearest integer]

Given: $R = 8.314\ J\ K^{-1}\ mol^{-1}$.

Open solution

Q15. Numerical

The elevation in boiling point for 1 molal solution of non-volatile solute A is $3\ K$. The depression in freezing point for 2 molal solution of A in the same solvent is 6 K. The ratio of K_b and K_f i.e., K_b/K_f is $1 : X$. The value of X is [nearest integer]

Open solution

Q16. Numerical

$20\ mL$ of $0.02\ M$ hypo solution is used for the titration of $10\ mL$ of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}$ M. [nearest integer]

Given : $2\,Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$

$I_2 + 2\,S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$

Open solution

Q17. Numerical

The spin-only magnetic moment value of the compound with strongest oxidizing ability among MnF_4, MnF_3 and MnF_2 is _____ B.M. [nearest integer]

Open solution

Q18. Numerical

Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is _____.

Open solution

Q19. Numerical

A $100\ mL$ solution of CH_3CH_2MgBr on treatment with methanol produces $2.24\ mL$ of a gas at STP. The weight of gas produced is _____ mg. [nearest integer]

Open solution

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Mathematics 20 questions

Q20. MCQ (Single Correct Answer)

The minimum value of the sum of the squares of the roots of $x^2 + (3-a)x + 1 = 2a$ is:

(A) 4 (B) 5 (C) 6 (D) 8

Open solution

Q21. MCQ (Single Correct Answer)

If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then :

(A) $x + 2y - 4 = 0$ (B) $x^2 + y - 4 = 0$ (C) $x + 2y + 4 = 0$ (D) $x^2 - y + 3 = 0$

Open solution

Q22. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of $A'BA$ is:

(A) 1224 (B) 1042 (C) 540 (D) 539

Open solution

Q23. MCQ (Single Correct Answer)

Let P and Q be any points on the curves $(x-1)^2 + (y+1)^2 = 1$ and $y = x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval :

(A) $\left(0, \frac{1}{4}\right)$ (B) $\left(\frac{1}{2}, \frac{3}{4}\right)$
(C) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{3}{4}, 1\right)$

Open solution

Q24. MCQ (Single Correct Answer)

If the maximum value of a, for which the function $f_a(x) = \tan^{-1} 2x - 3ax + 7$ is non-decreasing in $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$, is $\bar{a}$, then $f_{\bar{a}}\left(\frac{\pi}{8}\right)$ is equal to :

(A) $8 - \frac{9\pi}{4(9+\pi^2)}$ (B) $8 - \frac{4\pi}{9(4+\pi^2)}$
(C) $8\left(\frac{1+\pi^2}{9+\pi^2}\right)$ (D) $8 - \frac{\pi}{4}$

Open solution

Q25. MCQ (Single Correct Answer)

Let $\beta = \lim\limits_{x \to 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in \mathbb{R}$. Then the value of $\alpha + \beta$ is :

(A) $\frac{14}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Open solution

Q26. MCQ (Single Correct Answer)

The value of $\log_e 2 \frac{d}{dx}(\log_{\cos x} \operatorname{cosec} x)$ at $x = \frac{\pi}{4}$ is

(A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) -4 (D) 4

Open solution

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Q27. MCQ (Single Correct Answer)

$\int_0^{20\pi} (|\sin x| + |\cos x|)^2 dx$ is equal to

(A) $10(\pi + 4)$ (B) $10(\pi + 2)$ (C) $20(\pi - 2)$ (D) $20(\pi + 2)$

Open solution

Q28. MCQ (Single Correct Answer)

Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$, $x \in (-1, 1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

(A) $\frac{\pi}{3} - \frac{1}{4}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Open solution

Q29. MCQ (Single Correct Answer)

Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is _____.

(A) 12 (B) 13 (C) 14 (D) 16

Open solution

Q30. MCQ (Single Correct Answer)

If the line $x - 1 = 0$ is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point :

(A) $(-2\sqrt{5}, 6)$ (B) $(-\sqrt{5}, 3)$ (C) $(\sqrt{5}, -2)$ (D) $(2\sqrt{5}, 3\sqrt{6})$

Open solution

Q31. MCQ (Single Correct Answer)

If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1}x}{\alpha} = \frac{\cos^{-1}x}{\beta}$, then the value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is :

(A) $4\sqrt{(1-x^2)}\,(1-2x^2)$ (B) $4x\sqrt{(1-x^2)}\,(1-2x^2)$

(C) $2x\sqrt{(1-x^2)}\,(1-4x^2)$ (D) $4\sqrt{(1-x^2)}\,(1-4x^2)$

Open solution

Q32. MCQ (Single Correct Answer)

The integral $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)} dx$ is equal to

(A) $\frac{1}{2}\log_e \left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right| + C$

(B) $\frac{1}{2}\log_e \left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)}\right| + C$

(C) $\log_e \left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}\right| + C$

(D) $\frac{1}{2}\log_e \left|\frac{\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}-\frac{\pi}{6}\right)}\right| + C$

Open solution

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Q33. MCQ (Single Correct Answer)

The area bounded by the curves $y = |x^2 - 1|$ and $y = 1$ is

(A) $\frac{2}{3}(\sqrt{2}+1)$ (B) $\frac{4}{3}(\sqrt{2}-1)$ (C) $2(\sqrt{2}-1)$ (D) $\frac{8}{3}(\sqrt{2}-1)$

Open solution

Q34. Numerical

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{3, 6, 7, 9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is _____.

Open solution

Q35. Numerical

Numbers are to be formed between 1000 and 3000 , which are divisible by 4 , using the digits $1, 2, 3, 4, 5$ and 6 without repetition of digits. Then the total number of such numbers is _____.

Open solution

Q36. Numerical

The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to _____.

Open solution

Q37. Numerical

Suppose $y = y(x)$ be the solution curve to the differential equation $\frac{dy}{dx} - y = 2 - e^{-x}$ such that $\lim\limits_{x\to\infty} y(x)$ is finite. If a and b are respectively the x - and y-intercepts of the tangent to the curve at $x = 0$, then the value of $a - 4b$ is equal to _____.

Open solution

Q38. Numerical

Different A.P.'s are constructed with the first term 100, the last term 199, and integral common differences. The sum of the common differences of all such A.P.'s having at least 3 terms and at most 33 terms is _____.

Open solution

Q39. Numerical

The number of matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \{-1, 0, 1, 2, 3, \ldots\ldots, 10\}$, such that $A = A^{-1}$, is _____.

Open solution

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Physics 26 questions

Q40. MCQ (Single Correct Answer)

Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be :

(A) $1 : \sqrt{2}$ (B) $\sqrt{2} : 1$ (C) $2 : \sqrt{3}$ (D) $\sqrt{3} : 2$

Open solution

Q41. MCQ (Single Correct Answer)

In a Vernier Calipers, 10 divisions of Vernier scale are equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4th Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to $1\ \text{mm}$. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6th Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :

(A) 3.02 cm (B) 3.06 cm (C) 3.10 cm (D) 3.20 cm

Open solution

Q42. MCQ (Single Correct Answer)

A ball of mass $0.15\ \text{kg}$ hits the wall with its initial speed of $12\ \text{ms}^{-1}$ and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is $100\ \text{N}$, calculate the time duration of the contact of ball with the wall.

(A) 0.018 s (B) 0.036 s (C) 0.009 s (D) 0.072 s

Open solution

Q43. MCQ (Single Correct Answer)

A body of mass $8\ \text{kg}$ and another of mass $2\ \text{kg}$ are moving with equal kinetic energy. The ratio of their respective momentum will be :

(A) 1 : 1 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

Open solution

Q44. MCQ (Single Correct Answer)

Two uniformly charged spherical conductors A and B of radii $5\ \text{mm}$ and $10\ \text{mm}$ are separated by a distance of $2\ \text{cm}$. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 1 : 4

Open solution

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Q45. MCQ (Single Correct Answer)

The oscillating magnetic field in a plane electromagnetic wave is given by

$B_y = 5 \times 10^{-6} \sin 1000\pi \left(5x - 4 \times 10^8 t\right) T$. The amplitude of electric field will be :

(A) $15 \times 10^2\ \mathrm{Vm}^{-1}$ (B) $5 \times 10^{-6}\ \mathrm{Vm}^{-1}$

(C) $16 \times 10^{12}\ \mathrm{Vm}^{-1}$ (D) $4 \times 10^2\ \mathrm{Vm}^{-1}$

Open solution

Q46. MCQ (Single Correct Answer)

Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8\ \mathrm{ms}^{-1}$ and $2.0 \times 10^8\ \mathrm{ms}^{-1}$ respectively. The critical angle between them is :

(A) $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$ (B) $\tan^{-1}\left(\frac{2}{3}\right)$

(C) $\cos^{-1}\left(\frac{3}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

Open solution

Q47. MCQ (Single Correct Answer)

A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be :

(Take radius of earth $= 6400\ \mathrm{km}$ and $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)

(A) 800 km (B) 1600 km (C) 2133 km (D) 4800 km

Open solution

Q48. MCQ (Single Correct Answer)

A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}(M' < M)$. The ratio of de Broglie wavelength of two parts will be :

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 2 : 3

Open solution

Q49. MCQ (Single Correct Answer)

An ice cube of dimensions $60\ \mathrm{cm} \times 50\ \mathrm{cm} \times 20\ \mathrm{cm}$ is placed in an insulation box of wall thickness $1\ \mathrm{cm}$. The box keeping the ice cube at $0^\circ\mathrm{C}$ of temperature is brought to a room of temperature $40^\circ\mathrm{C}$. The rate of melting of ice is approximately :

(Latent heat of fusion of ice is $3.4 \times 10^5\ \mathrm{J\,kg}^{-1}$ and thermal conducting of insulation wall is $0.05\ \mathrm{Wm}^{-1\circ}\mathrm{C}^{-1}$)

(A) $61 \times 10^{-3}\ \mathrm{kg\,s}^{-1}$ (B) $61 \times 10^{-5}\ \mathrm{kg\,s}^{-1}$

(C) $208\ \mathrm{kg\,s}^{-1}$ (D) $30 \times 10^{-5}\ \mathrm{kg\,s}^{-1}$

Open solution

Q50. MCQ (Single Correct Answer)

A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

(A) $\frac{n}{n+2}$ (B) $\frac{n+2}{n}$ (C) $\frac{n}{2n+2}$ (D) $\frac{n}{n-2}$

Open solution

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Q51. MCQ (Single Correct Answer)

A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be :

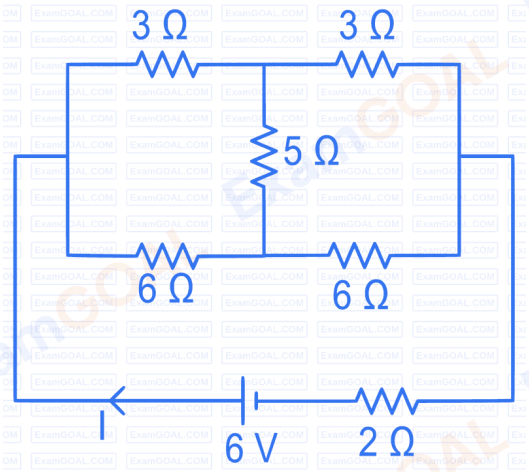
Open solution

(A) 4π (B) 2π (C) π (D) 2

Q52. MCQ (Single Correct Answer)

A battery of 6 V is connected to the circuit as shown below. The current I drawn from the battery is :

Open solution

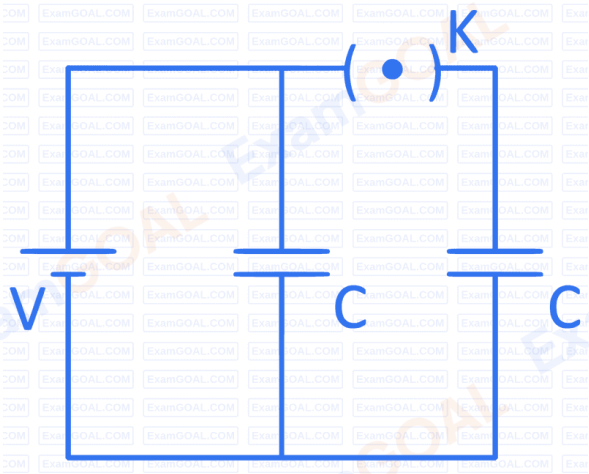


(A) 1A (B) 2A (C) $\frac{6}{11}$ A (D) $\frac{4}{3}$ A

Q53. MCQ (Single Correct Answer)

A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key 'K' is closed, the total energy stored across the combination is E_1. Now key 'K' is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2. The ratio E_1/E_2 will be :

Open solution



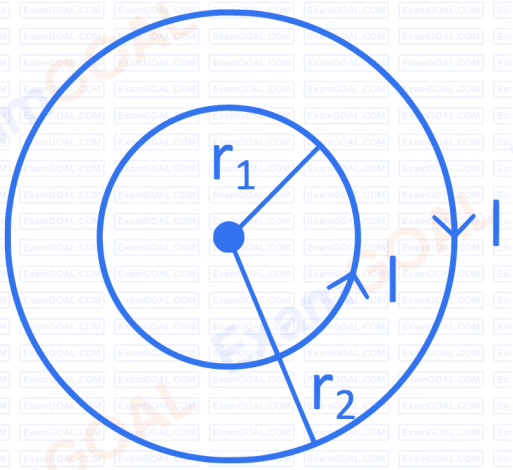
(A) $\frac{1}{10}$ (B) $\frac{2}{5}$ (C) $\frac{5}{13}$ (D) $\frac{5}{26}$

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Q54. MCQ (Single Correct Answer)

Two concentric circular loops of radii $r_1 = 30\ \text{cm}$ and $r_2 = 50\ \text{cm}$ are placed in $\text{X} - \text{Y}$ plane as shown in the figure. A current $I = 7\ \text{A}$ is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :

Open solution



(A) $\frac{7}{2}\hat{k}\,\text{Am}^2$

(B) $-\frac{7}{2}\hat{k}\,\text{Am}^2$

(C) $7\,\hat{k}\,\text{Am}^2$

(D) $-7\,\hat{k}\,\text{Am}^2$

Q55. MCQ (Single Correct Answer)

A velocity selector consists of electric field $\vec{E} = E\,\hat{k}$ and magnetic field $\vec{B} = B\,\hat{j}$ with $B = 12\,mT$. The value of E required for an electron of energy $728\,\text{e}V$ moving along the positive x-axis to pass undeflected is :

(Given, mass of electron $= 9.1 \times 10^{-31}\ \text{kg}$)

Open solution

(A) $192\,\text{kVm}^{-1}$

(B) $192\,\text{mVm}^{-1}$

(C) $9600\,\text{kVm}^{-1}$

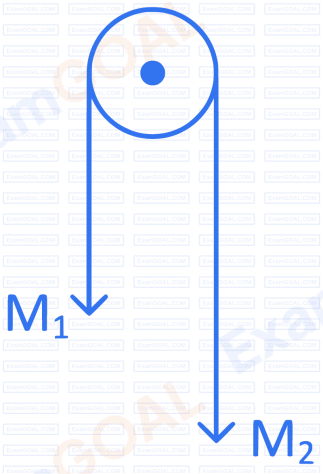
(D) $16\,\text{kVm}^{-1}$

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Q56. MCQ (Single Correct Answer)

Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1, the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1, the acceleration of the system is a_2. The ratio $\frac{a_1}{a_2}$ will be :

Open solution



(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{1}{2}$

Q57. MCQ (Single Correct Answer)

The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4}\,\mathrm{m}^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be :

(take $g = 10\,ms^{-2}$)

Open solution

(A) $6.25 \times 10^{-4}\,\mathrm{m}^2$ (B) $10 \times 10^{-4}\,\mathrm{m}^2$

(C) $1 \times 10^{-4}\,\mathrm{m}^2$ (D) $1.67 \times 10^{-4}\,\mathrm{m}^2$

Q58. Numerical

If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})\,\mathrm{m}$ and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})\,\mathrm{m}$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.

Open solution

Q59. Numerical

The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m.

Given, the length of the rod is $10\sqrt{3}\,\mathrm{m}$.

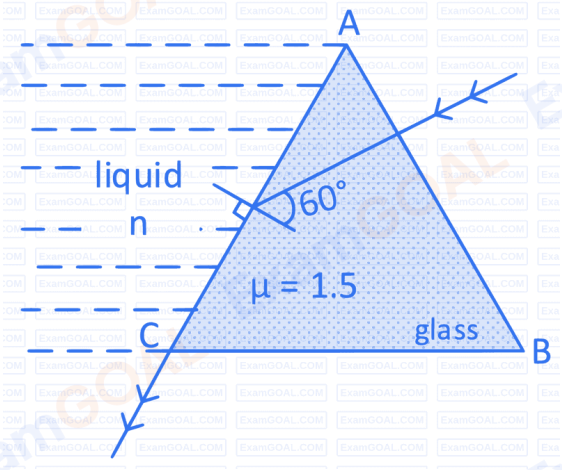
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Q60. Numerical

In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is _____.

(Given refractive index of glass $= 1.5$)



Open solution

Q61. Numerical

Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below :

${}_1^2X + {}_1^2X = {}_2^4Y$

The binding energies per nucleon for ${}_1^2X$ and ${}_2^4Y$ are $1.1\,\text{MeV}$ and $7.6\,\text{MeV}$ respectively. The energy released in this process is _____ MeV.

Open solution

Q62. Numerical

A uniform heavy rod of mass $20\,\text{kg}$, cross sectional area $0.4\,\text{m}^2$ and length $20\,\text{m}$ is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9}\,\text{m}$. The value of x is _____.

(Given, young modulus Y = 2×10^{11} Nm^{-2} and g = 10 ms^{-2})

Open solution

Q63. Numerical

Three point charges of magnitude $5\mu\text{C}, 0.16\mu\text{C}$ and $0.3\mu\text{C}$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3\,\text{cm}, BC = 3\sqrt{2}\,\text{cm}$ and $CA = 3\,\text{cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

Open solution

Q64. Numerical

In a coil of resistance $8\,\Omega$, the magnetic flux due to an external magnetic field varies with time as $\phi = \frac{2}{3}\left(9 - t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be _____ J.

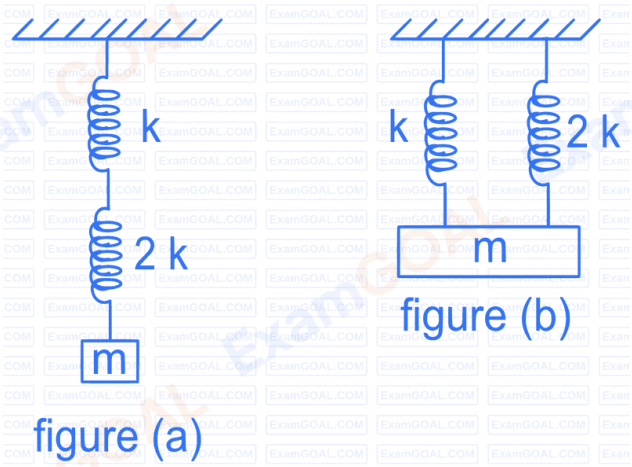
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Q65. Numerical

As per given figures, two springs of spring constants k and $2k$ are connected to mass m. If the period of oscillation in figure (a) is 3s, then the period of oscillation in figure (b) will be $\sqrt{x}\ s$. The value of x is _____.

Open solution



Out of syllabus 25 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q66. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Finest gold sol is red in colour, as the size of the particles increases, it appears purple then blue and finally gold.

Reason R : The colour of the colloidal solution depends on the wavelength of light scattered by the dispersed particles.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q67. MCQ (Single Correct Answer) · **Chemistry**

The metal that has very low melting point and its periodic position is closer to a metalloid is :

(A) Al (B) Ga (C) Se (D) In

Open solution

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Q68. MCQ (Single Correct Answer) · **Chemistry**

The metal that is not extracted from its sulfide ore is :

(A) Aluminium (B) Iron (C) Lead (D) Zinc

Open solution

Q69. MCQ (Single Correct Answer) · **Chemistry**

The products obtained from a reaction of hydrogen peroxide and acidified potassium permanganate are :

(A) Mn^{4+}, H_2O only (B) Mn^{2+}, H_2O only

(C) Mn^{4+}, H_2O, O_2 only (D) Mn^{2+}, H_2O, O_2 only

Open solution

Q70. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: LiF is sparingly soluble in water.

Reason R: The ionic radius of Li^+ ion is smallest among its group members, hence has least hydration enthalpy.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion : Boric acid is a weak acid.

Reason R : Boric acid is not able to release H^+ ion on its own. It receives OH^- ion from water and releases H^+ ion.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

Open solution

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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II.

	List I Pollutant		List II Source
(A)	Microorganisms	(I)	Strip mining
(B)	Plant nutrients	(II)	Domestic sewage
(C)	Toxic heavy metals	(III)	Chemical fertilizer
(D)	Sediment	(IV)	Chemical factory

Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I

(B) A-II, B-I, C-IV D-III

(C) A-I, B-IV, C-II, D-III

(D) A-I, B-IV, C-III, D-II

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Vulcanization of rubber is carried out by heating a mixture of

(A) isoprene and styrene

(B) neoprene and sulphur

(C) isoprene and sulphur

(D) neoprene and styrene

Q74. Numerical · **Chemistry**

Open solution

A $10\ \text{g}$ mixture of hydrogen and helium is contained in a vessel of capacity $0.0125\ \text{m}^3$ at 6 bar and 27°C. The mass of helium in the mixture is _____ g. (nearest integer)

Given: $\text{R} = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$

(Atomic masses of H and He are 1u and 4u, respectively)

Q75. Numerical · **Chemistry**

Open solution

The number of non-ionisable protons present in the product B obtained from the following reactions is _____.

$$\text{C}_2\text{H}_5\text{OH} + \text{PCl}_3 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{A}$$

$$\text{A} + \text{PCl}_3 \rightarrow \text{B}$$

Q76. Numerical · **Chemistry**

Open solution

How many of the following drugs is/are example(s) of broad spectrum antibiotics?

Ofloxacin, Penicillin G, Terpineol, Salvarsan.

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Q77. MCQ (Single Correct Answer) · **Mathematics**

$\sum\limits_{\substack{i,j=0 \\ i\neq j}}^{n} {}^{n}C_i \, {}^{n}C_j$ is equal to

(A) $2^{2n} - {}^{2n}C_n$

(B) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(C) $2^{2n} - \frac{1}{2}{}^{2n}C_n$

(D) $2^{2n-1} + {}^{2n-1}C_n$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point $(1, 3)$ is :

(A) $\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$

(B) $\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

(C) $\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$

(D) $\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x-2)^2$ is

(A) $y = 4(x-2)$

(B) $y = 4(x-1)$

(C) $y = 4(x+1)$

(D) $y = 4(x+2)$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is :

(A) $\frac{3\pi}{4}$

(B) $\frac{2\pi}{3}$

(C) $\frac{4\pi}{5}$

(D) $\frac{5\pi}{6}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Negation of the Boolean expression $p \Leftrightarrow (q \Rightarrow p)$ is

(A) $(\sim p) \wedge q$

(B) $p \wedge (\sim q)$

(C) $(\sim p) \vee (\sim q)$

(D) $(\sim p) \wedge (\sim q)$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$. Then, $54\,P(X \leq 2)$ is equal to :

(A) $\frac{73}{27}$

(B) $\frac{146}{27}$

(C) $\frac{146}{81}$

(D) $\frac{126}{81}$

Open solution

Q83. Numerical · **Mathematics**

The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(\hat{i} + a\hat{j} - \hat{k})$ and $\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - a\hat{k})$ from the point $(2, 1, 4)$ is $\sqrt{3}$, is _____.

Open solution

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Q84. Numerical · **Mathematics**

If $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$, where m and n are co-prime, then $m+n$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

The plane passing through the line $L : lx - y + 3(1-l)z = 1, x + 2y - z = 2$ and perpendicular to the plane $3x + 2y + z = 6$ is $3x - 8y + 7z = 4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

The maximum and minimum voltage of an amplitude modulated signal are 60 V and 20 V respectively. The percentage modulation index will be :

(A) 0.5% (B) 50% (C) 2% (D) 30%

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Mass numbers of two nuclei are in the ratio of $4:3$. Their nuclear densities will be in the ratio of

(A) 4 : 3 (B) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$

(C) 1 : 1 (D) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

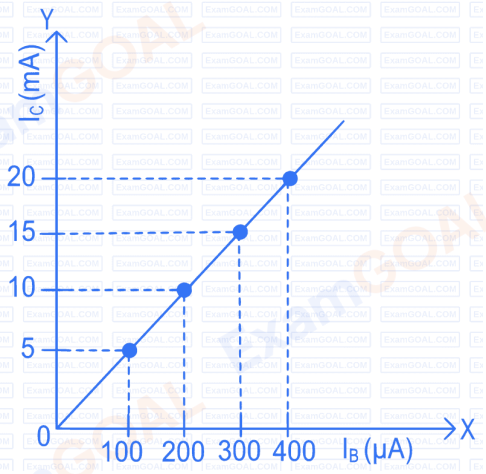
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Q89. Numerical · **Physics**

The typical transfer characteristics of a transistor in CE configuration is shown in figure. A load resistor of $2\,k\Omega$ is connected in the collector branch of the circuit used. The input resistance of the transistor is $0.50\,\mathrm{k}\Omega$. The voltage gain of the transistor is _____.

Open solution



Q90. Numerical · **Physics**

A potentiometer wire of length $300\,\mathrm{cm}$ is connected in series with a resistance 780 Ω and a standard cell of emf $4\mathrm{V}$. A constant current flows through potentiometer wire. The length of the null point for cell of emf $20\,\mathrm{mV}$ is found to be $60\,\mathrm{cm}$. The resistance of the potentiometer wire is _____ Ω.

Open solution

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Answer Key

Chemistry

1. **C**	2. **B**	3. **C**	4. **C**	5. **A**
6. **B**	7. **A, B**	8. **B**	9. **A**	10. **B**
11. **C**	12. **C**	13. **4**	14. **57**	15. **1**
16. **4**	17. **5**	18. **12**	19. **3**	

Mathematics

20. **C**	21. **C**	22. **D**	23. **C**	24. **A**
25. **C**	26. **D**	27. **D**	28. **B**	29. **A**
30. **C**	31. **B**	32. **A**	33. **D**	34. **112**
35. **30**	36. **238**	37. **3**	38. **53**	39. **50**

Physics

40. **C**	41. **C**	42. **B**	43. **B**	44. **B**
45. **D**	46. **A**	47. **A**	48. **C**	49. **B**
50. **A**	51. **A**	52. **A**	53. **C**	54. **B**
55. **A**	56. **B**	57. **A**	58. **2**	59. **5**
60. **27**	61. **26**	62. **25**	63. **17**	64. **2**
65. **2**				

Out of syllabus

66. **A**	67. **B**	68. **A**	69. **D**	70. **C**
71. **A**	72. **A**	73. **C**	74. **8**	75. **2**
76. **1**	77. **A**	78. **B**	79. **B**	80. **A**
81. **D**	82. **B**	83. **2**	84. **166**	85. **3**
86. **125**	87. **B**	88. **C**	89. **200**	90. **20**

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